



Practical Pharmacology-3 (Pharmacokinetics-Absorption)

Quiz no. two: 10 marks (4 questions)

(Clinical scenario and relevance: 2 marks, pharmacological questions: 4 marks,

patient's guide: 3 marks, correct spelling of drug names: 1 mark)

Clinical problem No. 6: A 25-year old female patient with repeated attacks of migraine headache, obtains no relief of her headache after taking two paracetamol (500 mg) tablets. On consulting a physician, he prescribed metoclopramide 10mg tablet to be taken half an hour before paracetamol. (مناقشة امتصاص الادوية من خلال حالة مريضة)		
Clinical relevance (Early exposure to clinical situations)	Try to look for a short description of the disease mentioned in this problem. (Answers are shown below)	What characterizes migraine's headache compared with other types of headache? Could migraine headache be bilateral? What might precede the attack of migraine headache? What relationship does it have to light, sound, physical activity, food, ...? What are the GI manifestations associated with migraine? Name one other disease that is associated with gastroparesis. (Answers are shown below)
Pharmacological questions (Answers are shown below)	Q1:	Which part of the GIT is the main part for drug absorption? The onset of absorption of acidic drugs occurs in the stomach; however, stomach is not considered the main site for absorption; why?
	Q2:	What are the factors that might affect drug absorption from the GIT? Define the term "bioavailability" and how can it be calculated?
	Q3:	Give examples of drugs that can increase or decrease gastric emptying. How can food affect drug absorption?
	Q4:	Students should have some early knowledge about the pharmacology of paracetamol and also metoclopramide. How does metoclopramide work in migraine? Why is it called a motility stimulant (prokinetic) drug? What are its main side effects?
	Q5:	How does facilitated diffusion differ from passive one and from active transport; give examples?
	Q6:	How can food affect drug absorption?
Patient Information (Patient's Guide)	What would you tell the patient about these drugs?	Paracetamol: - Dose correctly: 500-1000mg every 4-6 hours. Never take more than 8 x 500mg tablets (4 grams) in one day. Taking more than this dose or taking this high dose for long duration can seriously damage the liver. - Look for "paracetamol" or "acetaminophen" on the labels of all other medicines. Many over-the-counter (OTC) cold, flu, sinus and joint medicines also contain paracetamol. A patient might take

		multiple products containing paracetamol and exceeding the dose without realizing it. 3. Avoid drinking alcohol while taking paracetamol (both can damage the liver). 4. You can take it with or without food. If it upsets your stomach, taking it with food might help. Metoclopramide 1. Used to relieve nausea and vomiting and to help your stomach empty faster. 2. It can be taken about 30 minutes before meals and at bedtime. 3. You must stop taking metoclopramide and seek immediate medical attention if you experience any of the following: involuntary movements, parkinsonism-like symptoms, or restlessness (akathisia). The risk of these movement disorders is higher in extremes of age; with longer use, and with higher total doses. 4. It can cause drowsiness (be cautious when driving or operating machinery) and diarrhea.
Drug formulations available will be shown in the lab: PARACETAMOL (ACETAMINOPHEN) 10mg/1ml IV Solution, 100mg/1ml drop, 120-125mg/5ml syrup, 120-250mg/5ml Suspension, 500mg tab, 125mg - 250mg suppository [Paracetamol combination with: vitamin C and chlorpheniramine maleate (Flu-out), with orphenadrine (Norgesic, myogesic, kanagesic), METOCLOPRAMIDE (Primperan): Tablets 10mg, syrup 5mg/5ml, injection 10mg/2ml , suppository 20mg (Some drug images and photos are shown below) (مطلوب من الطالب معرفة الاشكال الدوائية التي يتوفر من خلالها الباراسيتامول والمتوكلوپراميد فقط – التي هي باللون الأحمر – أما الاسماء التجارية والجرعات غير مطلوبة)		
Summative assessment (Drug names to be written without any spelling mistake are: paracetamol (acetaminophen), metoclopramide		

Brief answers of the clinical questions

1. What characterizes migraine's headache compared with other types of headache?

Migraine headache is distinct from other common headaches like tension-type headache or cluster headache in several key ways:

- **Quality of Pain:** The pain is typically **throbbing or pulsating**, in contrast to the constant, dull, pressure-like ache of tension headache. It is usually unilateral but can be bilateral.
- **Intensity:** The pain is usually **moderate to severe**, often inhibiting or preventing daily activities.
- **Duration:** Attacks last anywhere from **4 to 72 hours** if untreated.
- **Associated Symptoms:** Migraine is more than just a headache. It is often accompanied by **nausea, vomiting, and/or sensitivity to light (photophobia) and sound (phonophobia)**. These are less common in tension headache.

- **Aggravation by Activity:** Routine physical activities (like walking up stairs) typically **aggravate** the pain, whereas they have little effect on tension headache.

2. Could migraine headache be bilateral?

Yes. While migraine is often thought of as a unilateral (one-sided) headache, it can be bilateral. In fact, the International Classification of Headache Disorders (ICHD-3) states that the headache can be unilateral **or** bilateral. Bilateral pain is particularly common in children and in certain subtypes of migraine.

3. What might precede the attack of migraine headache?

For about 25-30% of individuals with migraine, the headache is preceded or accompanied by a **neurological disturbance known as "aura."**

Aura symptoms typically develop gradually over 5-20 minutes and last less than 60 minutes. They can include:

- **Visual disturbances:** Most common (e.g., seeing flickering lights, zigzag lines, blind spots, or loss of part of the visual field).
- **Sensory disturbances:** "Pins and needles" sensation or numbness that travels up one limb.
- **Speech disturbances:** Difficulty speaking or finding words.

Some people also experience **prodromal symptoms** hours or even a day or two before the headache begins. These are not the same as an aura and can include:

- Mood changes (e.g., depression, euphoria, irritability)
- Fatigue or yawning excessively
- Food cravings or loss of appetite
- Neck stiffness
- Increased urination

4. What relationship does it have to light, sound, physical activity, food, ...?

Migraine has a strong relationship with various internal and external factors:

- **Light (Photophobia) & Sound (Phonophobia):** These are hallmark symptoms. During an attack, normal levels of light and sound can become intolerable and worsen the pain. Patients often seek out a dark, quiet room.
- **Physical Activity:** As mentioned, routine physical activity **aggravates** the headache.

- **Food:** Certain foods are common triggers for some people (but not all), including aged cheeses, processed meats (nitrates), chocolate, monosodium glutamate, and alcohol (especially red wine).
- **Other Common Triggers:** Hormonal changes in women, stress, changes in sleep pattern, skipping meals, strong smells, and weather changes.

5. What are the GI manifestations associated with migraine?

Gastrointestinal (GI) manifestations are a core feature of migraine, not just a secondary effect of pain. They include:

- **Nausea** (the most common)
- **Vomiting**
- **Abdominal pain** or discomfort
- **Loss of appetite**
- **Diarrhea** (less common)
- **Gastroparesis** - a delayed emptying of the stomach's contents into the small intestine, which can impair the absorption of oral medications.

6. Name one other disease that is associated with gastroparesis.

Diabetes Mellitus is a very common disease associated with gastroparesis. Long-term high blood sugar can damage the vagus nerve, which controls the movement of food through the digestive tract, leading to diabetic gastroparesis.

Brief answers of the pharmacological questions

Q1: Drug Absorption in the GIT

- **Which part of the GIT is the main part for drug absorption?**
The **small intestine**, specifically the duodenum and jejunum, is the primary site for drug absorption.
- **The onset of absorption of acidic drugs occurs in the stomach; however, the stomach is not considered the main site for absorption even for acidic drugs; why?**
While the stomach's acidic environment favors the unionized (absorbable) form of acidic drugs, its surface area is very small compared to the small intestine. The small intestine has a very large surface area due to the presence of villi and microvilli, which is the most important factor for efficient absorption. Furthermore, the transit time of drugs in the stomach is often short (i.e. rapid gastric emptying), while the intestinal transit time is longer and more consistent, allowing for more complete absorption.

Q2: Factors Affecting Drug Absorption & Bioavailability

- **Factors affecting drug absorption from the GIT:**
 - **Physicochemical Properties of the Drug:** Lipid solubility, degree of ionization (pKa and pH), molecular size, and dissolution rate.
 - **Patient & Physiological Factors:** Gastrointestinal pH, surface area, blood flow to the absorption site, gastric emptying time, intestinal motility, and presence of food.
 - **Dosage Form Factors:** Solution, tablet, capsule, sustained-release formulations.
 - **First-Pass Metabolism:** Metabolism of the drug in the liver and gut wall before it reaches systemic circulation.
- **Define the term “bioavailability” and how can it be calculated?**
 - **Definition:** Bioavailability (F) is the fraction (or percentage) of an administered drug that reaches the systemic circulation unchanged (intact) and becomes available at the site of action.
 - **Calculation:** It is most commonly calculated by comparing the Area Under the Curve (AUC) after oral administration to AUC after intravenous (IV) administration (which has 100% bioavailability by definition).
Formula: $F = (AUC_{oral} / AUC_{IV})$ (AUC; the curve is the plasma concentration versus time curve).

Q3: Drugs Affecting Gastric Emptying

- **Drugs that Increase Gastric Emptying (Prokinetics):**
 - Metoclopramide
 - Domperidone
 - Erythromycin (a macrolide antibiotic with prokinetic effects)
- **Drugs that Decrease Gastric Emptying (Delay Emptying): Examples:**
 - Opioids (e.g., Morphine, Codeine)
 - Anticholinergics (e.g., Atropine, ...)
 - Tricyclic Antidepressants (e.g., Amitriptyline)

Q4: Pharmacology of Paracetamol & Metoclopramide

- **Paracetamol (Acetaminophen):**
 - **Class:** Centrally-acting analgesic and antipyretic (reduces somatic and fever).
 - **Mechanism:** It is believed to inhibit cyclooxygenase (COX) enzymes, particularly a variant called COX-3, within the central nervous system. This reduces the production of prostaglandins, which are mediators of pain and fever. It has minimal anti-inflammatory effects peripherally.
- **Metoclopramide in Migraine:**
 - **How it works:** Metoclopramide works in migraine through a **dual mechanism**:
 1. **Dopamine (D2) Receptor Antagonism:** By blocking dopamine receptors in the chemoreceptor trigger zone (CTZ) in the brain, it provides **anti-nausea and anti-emetic** effects, which are useful for treating migraine-associated nausea and vomiting.
 2. **Prokinetic Effect:** It increases gastric emptying and upper GI motility. This is critically important in migraine, as migraine often causes **gastroparesis** (slowed stomach emptying). By speeding up gastric emptying, it ensures that co-administered oral pain relievers (like paracetamol or NSAIDs) are moved from the stagnant **خامل** stomach into the small intestine for proper absorption, making them more effective.
- **Why is it called a motility stimulant (prokinetic) drug?**

It is called a prokinetic because it **stimulates motility (movement) of the upper gastrointestinal tract**. It does this by:

 - Antagonizing inhibitory dopamine receptors in the gut (Dopamine inhibits Ach release).
 - Potentiating the action of acetylcholine (the primary neurotransmitter for gut motility) at muscarinic receptors (It is not muscarinic agonist, but causes the release of Ach; this Ach stimulates muscarinic receptors in the gut i.e. indirectly acting).
- **Main Side Effects of Metoclopramide:**
 - **Extrapyramidal Symptoms (EPS):** Due to its central D2 blockade, it can cause dystonia (muscle spasms), parkinsonism (tremor, rigidity), akathisia (motor restlessness), and others.
 - **Drowsiness and fatigue**
 - **Hyperprolactinemia:** Can cause galactorrhea (milk discharge) and menstrual irregularities and infertility due to increased prolactin secretion.

Q5: Transport Mechanisms of drugs across cell membranes

Feature	Passive Diffusion	Facilitated Diffusion	Active Transport
Definition	Movement down a concentration gradient (from high to low) without a carrier.	Movement down a concentration gradient with a carrier protein.	Movement against a concentration gradient with a carrier protein, requiring energy (ATP).
Energy Required	No	No	Yes
Saturation	No	Yes (saturable)	Yes (saturable)
Selectivity	Low	High	High
Example	Most drugs (e.g., Paracetamol).	Glucose into most cells via GLUT transporters.	Levodopa into the brain via amino acid transporters.

Q6: How can food affect drug absorption?

1. Food can **DECREASE** absorption (may be taken on an empty stomach) "**Empty stomach**" means **1 hour or more before a meal or 2 hours or more after a meal**". Examples: Antibiotics: e.g. ampicillin, tetracyclines (tetracyclines can chelate with $\text{Ca}^{2+}/\text{Mg}^{2+}/\text{Al}^{3+}$ in, for example, milk and antacids → insoluble complexes and reduced bioavailability), and many other examples.

2. Food can **INCREASE** absorption (may be taken with food). This may improve drug effectiveness. Examples: Fat-soluble drugs: Griseofulvin (antifungal), Isotretinoin (vitamin A derivative used for acne), and others. Taking these with a high-fat meal can double or triple their absorption.

Grapefruit Juice: This is a common and potentially dangerous interaction between food and drugs. Grapefruit juice **inhibits** an enzyme in the intestine (CYP3A4) that breaks down many drugs. This can lead to high levels of the drug in bloods and may cause toxicity. Examples of affected Drugs: Some Statins (Atorvastatin, Simvastatin: lipid lowering), some blood pressure medications (e.g. Felodipine, nifedipine), and many others. Avoid grapefruit juice completely if you are on a medication with this known interaction.





Elixir

100 ml





Health And Beauty



